

RAJEEV POLEPALLI

Python Full Stack Developer

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PROFESSIONAL SUMMARY:

Experienced **Python Full-Stack Developer** (10+ years) building scalable, cloud-native web applications using **Flask, Django, React.js** (Hooks, Redux, Redux Toolkit), **Vue.js**, and **Node.js** (Express). Proficient in designing **RESTful APIs**, deploying on **AWS**, and working with **MySQL** and **MongoDB**. Skilled in **DevOps** practices and **CI/CD pipelines** using **Jenkins, GitHub Actions**, and **Docker** for reliable, automated deployments.

- Over **10 years** of experience in software development, including analysis, design, development, deployment, and maintenance of scalable client-server, web-based, and distributed applications.
- Proven expertise in full-stack development using **Python** and modern frontend frameworks, with strong exposure to **DevOps** and **cloud** technologies.
- Extensive experience with **Python** and **object-oriented** programming, including **multithreading**, **exception handling**, and adherence to **PEP-8** best practices.
- Proficient in web frameworks such as **Django, Wheezy.web, FastAPI** and **Flask**, experience with building RESTful APIs, backend services, and microservices.
- Hands-on experience with Python libraries like requests, grequests, urllib, BeautifulSoup, celery, jinja2, itertools, lxml, pandas, numpy, matplotlib.
- Experience with ORM frameworks such as SQLAlchemy, strong in SQL development including stored procedures and query optimization.
- Full-stack development experience with **JavaScript, TypeScript, Vue.js, React.js, JQuery, HTML5, CSS3**, and responsive UI frameworks such as **Bootstrap**.
- Developed progressive web applications (**PWAs**) with offline-first capabilities and service worker integration.
- Hands-on experience with **MongoDB** for designing and querying NoSQL document-based databases, proficient in schema design, aggregation framework, and indexing strategies.
- Worked with **PostgreSQL, MySQL, Oracle**, and **DynamoDB**, leveraging both relational and NoSQL models based on application needs.
- Extensive experience using **Pytest** for unit testing, integration testing, and test-driven development (TDD).
- Deep hands-on experience with **Amazon Web Services (AWS)**: EC2, S3, RDS, DynamoDB, ECS, Lambda, SQS, SNS, API Gateway, Kinesis, Glue, Athena, EMR, and Redshift.
- Built and deployed cloud-native applications using **Docker, Kubernetes, Terraform**, and **Packer**.
- Setup and managed **CI/CD pipelines** using **Jenkins** for automated builds, testing, and deployments.
- Proficient in configuring and optimizing **Gunicorn, WSGI**, and **ASGI** web servers.
- Experience deploying applications on **Apache**, and **Nginx**.
- Implemented caching strategies using **Redis** for high-performance data retrieval.
- Strong background in data querying, transformation, and analytics using **PySpark, Pandas, NumPy**, and cloud-native tools like **AWS Glue, Amazon Athena, EMR, Redshift**, and **Kinesis**.
- Experience in developing data pipelines and real-time data processing solutions.
- Skilled in version control using **Git, Bitbucket** and **TFS**.
- Familiar with **Agile** and Waterfall methodologies, and experienced in delivering high-quality solutions on time across multiple projects.
- Proficient in using development and project management tools including VS Code, PyCharm, Eclipse, Sublime Text, JIRA, and Salesforce.
- **Led** and **mentored** junior developers, providing guidance on code quality, architecture, and best practices.

EDUCATION:

- Master of Science in Computer Science, 2014(Texas A&M University - Kingsville, Texas, USA)
- Bachelor of Technology (B.Tech), Computer Science, 2012 (JNTUH, India)

TECHNICAL SKILLS:

- **Programming Languages:** Python, Java (J2SE), C
- **Framework:** Flask, Wheezy Web, Django, Node.js, FastAPI
- **Front End:** HTML, CSS, JavaScript, TypeScript, Bootstrap, Vue.js, ReactJS, NextJS
- **Database:** SQL Server, My SQL, SQLite, DynamoDB, MongoDB
- **Data & Analytics:** Pandas, NumPy, PySpark, AWS Athena, EMR, Redshift, Kinesis, Elasticsearch
- **Cloud Platform:** AWS (EC2, S3, Lambda, RDS, SQS, SNS, API Gateway), Google Cloud Platform (GCP)
- **DevOps & Containers:** Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Packer
- **Testing & Automation:** Pytest, unittest, Postman, Test-Driven Development (TDD), CI/CD pipelines
- **Version Control:** Git, Bitbucket, TFS
- **Ticketing Tools:** JIRA, Salesforce
- **IDEs:** VS Code, PyCharm, Eclipse, Sublime Text
- **SDLC:** Agile, Waterfall

PROFESSIONAL EXPERIENCE:

ELEVANCE HEALTH, ATLANTA, GA

JUL 2025 – PRESENT

ROLE: PYTHON FULL STACK DEVELOPER

RESPONSIBILITIES:

- Developed secure **RESTful APIs** using **Flask** for reliable access and management of sensitive healthcare data, supporting key use cases including provider information lookup, claims queries, and audit tracking.
- Maintained and enhanced a **React**-based portal by refactoring components, updating **Redux** state management, and improving the usability of provider and claims modules.
- Improved portal responsiveness and reduced redundant API calls by leveraging **React memoization**, implementing component-level caching, and optimizing data-fetching patterns.
- Contributed to **ETL** workflows using Python, **Pandas**, and **Numpy** to clean, transform, and validate healthcare provider data for accurate reporting and integration with the internal portal.
- Contributed to **ETL** workflows using Python, **PySpark**, **Pandas**, and **Numpy** to clean, transform, and validate healthcare provider data for accurate reporting and integration with the internal portal.
- Leveraged **PySpark's** distributed data processing capabilities to efficiently scale data transformations and aggregations, improving the performance and scalability of the ETL pipeline for large healthcare datasets.
- Implemented **unit** and **integration tests** for Python backend APIs and ETL jobs using **Pytest**, improving code quality and supporting team-wide **CI/CD**.
- Utilized **MySQL** for managing and querying processed provider data, ensuring reliable data storage and efficient access for reporting and internal portal functionalities.
- Implemented caching strategies using **Redis** to improve data retrieval performance and reduce backend load, enhancing responsiveness for different modules in the internal portal.
- Implemented **Docker**-based containerization for Python ETL scripts and backend APIs, enabling consistent deployments and simplified environment setup across development and production.
- Leveraged AWS services such as **S3**, **Lambda**, and **EC2** to automate secure storage, processing, and hosting of healthcare provider data. Utilized **API Gateway** for scalable API delivery and **SQS** for managing asynchronous job workflows supporting the internal portal.
- Collaborated with cross-functional teams in an **Agile** environment, participating in sprint planning, daily stand-ups, and task tracking using **JIRA** to ensure timely delivery of project enhancements.

ENVIRONMENT: Python, Flask, MySQL, ReactJS, PySpark, GIT, AWS, Redis, HTML, CSS, JavaScript, TypeScript, Docker, Visual Studio Code, Agile Methodology

CORAL INNOVATIONS

MAY 2024 – JUN 2025

ROLE: SENIOR SOFTWARE ENGINEER

RESPONSIBILITIES:

- Designed and developed high-performance backend APIs using **Python (Flask)** for machine learning inference and **Node.js** for portal services.
- Architected and deployed RESTful services through **AWS Lambda, API Gateway, and EC2**, ensuring scalability and minimal latency.
- Integrated **MongoDB** for dynamic data storage and optimized queries for fast retrieval in analytics-heavy components.
- Developed dynamic user dashboards and interactive UI components using **ReactJS**, enhancing data accessibility and user workflows.
- Configured **AWS S3** for secure, scalable storage of structured/unstructured data, implemented cost-optimized retrieval and archival strategies.
- Integrated an Amazon S3-based Data Lake to store and organize raw and processed data, enabling scalable storage and analytics for large datasets ingested from microservices and user activity logs.
- Leveraged **Redis** as a caching layer for performance-critical API endpoints and asynchronous task queues.
- Deployed ML model endpoints using **Flask-based APIs** on EC2 with **Gunicorn** and **NGINX** for production-grade inference serving.
- Managed CI/CD pipelines using **Jenkins**, automating build and deployment workflows.
- Created and maintained **custom AMIs** and **Launch Configurations** for reliable and repeatable infrastructure scaling.
- Enabled **Auto Scaling Groups** to handle high traffic scenarios while optimizing cloud cost and ensuring system availability.
- Set up **CloudWatch alarms**, logs, and dashboards for end-to-end monitoring, performance tuning, and proactive incident handling.
- Consumed and orchestrated background tasks using **AWS SQS**, ensuring decoupling and resiliency in asynchronous workflows.
- Configured **Nginx reverse proxy rules** for secure routing of internal APIs, static content caching, and request throttling.
- Applied **JWT and OAuth2** standards for token-based authentication, securing both Flask and Node.js microservices.
- Scheduled periodic data processing jobs using **cron on EC2 and AWS CloudWatch Events**, optimizing resource utilization.
- **Processed and analyzed over 3 crore user records monthly** using Python (Pandas, NumPy), enabling usage trend insights and behavioral analytics.
- Conducted **code reviews**, enforced clean code and security practices, and mentored junior developers across full-stack tasks.
- Participated actively in **Agile sprints**, sprint planning, retrospectives, and client demos while ensuring timely delivery of milestones.

ENVIRONMENT: Python, Flask, Node.js, MongoDB, ReactJS, GIT, AWS, Redis, HTML, CSS, JavaScript, TypeScript, Docker, Linux, Visual Studio Code, Agile Methodology

CLIENT: INCUS5 INC, AZ, USA (INDIA)

JAN 2020 – MAY 2024

ROLE: LEAD CONSULTANT

ROLES & RESPONSIBILITIES:

- Designed and developed scalable RESTful APIs using **Django REST Framework** for core business functionalities.
- Implemented **microservices architecture**, enabling independently deployable services with clear boundaries and API contracts.
- Developed dynamic, component-driven frontend UIs using **Vue.js**, integrating with REST APIs via Axios and handling state using Vuex.
- Integrated secure SSO using **OAuth2/OpenID Connect** via OKTA, Microsoft, and Google Identity providers.

- Deployed containerized microservices using **Docker** and orchestrated environments using **Docker Compose** for local and staging setups.
- Defined **JWT-based authentication**, built middleware for token validation, and enforced **RBAC (Role-Based Access Control)** across services.
- Developed scalable MySQL schemas, optimized query performance using indexes, normalized structures, and proper data types.
- Created reusable Django ViewSets, Serializers, and routers to streamline backend CRUD and custom business logic endpoints.
- Designed and implemented **GraphQL** APIs to enable clients to query exactly the data they need, minimizing over-fetching and enhancing network efficiency.
- Implemented multi-threading and concurrency to efficiently schedule and execute batch jobs, improving application performance and resource utilization.
- Implemented **unit, integration, and contract testing** using **Pytest** and shared schemas to validate service interactions.
- Developed reusable UI components and route-based lazy loading in Vue.js, optimized for performance using Webpack.
- Leveraged **Swagger/OpenAPI** for documenting and testing APIs, and shared collections via Postman for QA and dev teams.
- Applied **CI/CD practices** using Git, Docker, and GCP pipelines to automate testing, builds, and deployments across microservices.

ENVIRONMENT: Python, DRF, Vue.js, ReactJS, Progressive Web Apps (PWA), GIT, Windows, GCP, Visual Studio Code, HTML, CSS, JavaScript, My SQL, GraphQL, Agile Methodology, Docker.

SAMOSA LABS SOFTWARE SOLUTIONS, INDIA

MAY 2017 – JAN 2020

ROLE: PYTHON DEVELOPER

RESPONSIBILITIES:

- Developed new backend endpoints for features such as comments, contacts, uploads, like/dislike, and follow/unfollow using **Python** and the **Wheezy Web** framework.
- Built a content management system supporting 10,000+ video uploads with scalable backend architecture.
- Integrated the application with **AWS** services (**S3, EC2, DynamoDB, SQS**) using Boto3, deployed containerized Python web services via **Docker**.
- Designed and implemented RESTful APIs to enable seamless interaction between frontend components and backend services.
- Enhanced internal dashboard using **ReactJS** and **Python** to monitor uploads, comments, flags, and user activity.
- Utilized **Redis** for session management and caching to improve application performance and user experience.
- Implemented asynchronous task handling for video processing and email notifications using **Celery** and **AWS SQS**.
- Integrated **Elasticsearch** for fast, full-text search capabilities across videos and user content.
- Automated infrastructure provisioning using **Terraform** and **Docker**, improved deployment consistency across environments.
- Built reusable frontend components with **ReactJS** to streamline development and maintain UI consistency.
- Performed database design and optimization in **DynamoDB** and **Elasticsearch** to reduce read/write latency.
- Configured logging and monitoring with AWS **CloudWatch** to track system performance and diagnose issues.
- Used Python modules such as **requests, grequests, BeautifulSoup** for **web crawling**.
- Wrote unit and integration tests using **Pytest**, ensured high test coverage and reliable feature delivery.
- Engaged in code reviews and Agile practices to promote knowledge sharing and drive continuous improvement.

ENVIRONMENT: Python, Wheezy Web, ReactJS, Redis, Amazon Web Services (AWS), HTML, CSS, JavaScript, DynamoDB, Elasticsearch, Ubuntu, PyCharm, Docker, Terraform, Packer.

CLIENT: Q2EBANKING, AUSTIN, TX, USA

AUG 2016 – APR 2017

ROLE: PYTHON DEVELOPER

RESPONSIBILITIES:

- Developed application extensions using a **custom Python framework**, working on features such as eStatement Enrollment, Loan Payment, Address Change, Alerts, and Marketing API.
- Created **Python utilities** to consume REST APIs, invoke stored procedures, and render dynamic outputs using templating.
- Transformed **XML-based inputs** into structured application logic, generating corresponding XML outputs for downstream systems.
- Designed and implemented **stored procedures** in **SQL Server** for efficient data access, insertion, and transformation.
- Developed responsive frontend components using **HTML**, **CSS**, and **JavaScript** to support user interaction features.
- Participated in **requirement gathering**, business logic analysis, and designing scalable solutions to meet client needs.
- Followed **Agile development methodology**, actively contributing to sprint planning, daily stand-ups, and retrospectives.
- Wrote and maintained **unit tests** to verify the integrity of backend logic and support regression testing for enhancements.
- Collaborated with QA and team members to **test, debug, and deploy** modules in production and UAT environments.
- Used **RabbitMQ** to handle background tasks asynchronously, improving application performance and allowing services to work independently.
- Managed source code and peer-reviewed changes using **Stash (Bitbucket)** to ensure clean commits and collaboration.

ENVIRONMENT: Python, Custom Framework, JSON, XML, SQL Server, HTML, CSS, JavaScript, JIRA, Stash, SQL Server Stored Procedures, Windows, Agile Methodology, PyCharm.

MERRILL CORPORATION, ST. PAUL, MN

MAY 2015 – JUL 2016

ROLE: PYTHON DEVELOPER

RESPONSIBILITIES:

- Developed web application modules using **Python**, **Django**, and **JSON**, following industry-standard coding practices.
- Configured Django settings to manage **URL routing**, middleware, application parameters, and static files.
- Used **Python** in generating **XML** based on the input data provided and also worked on **project enhancements**.
- Built and validated XML skeletons using provided **XSD schemas**, ensuring data structure compliance.
- Developed Django **views and templates** to create a user-friendly and responsive web interface.
- Accessed and managed databases using **Django ORM**, performing complex queries on **SQLite** and **MySQL**.
- Built UI components using **HTML5**, **CSS3**, **JavaScript**, and **AngularJS**, contributing to responsive frontend development.
- Read structured data from **Excel**, processed it, and exported reports into **CSV** format.
- Designed and implemented **data visualizations** using **matplotlib**, including dynamic chart generation from Excel inputs.
- Developed and consumed **SOAP** web services to facilitate data exchange between distributed systems.
- Developed the required **XML Schema** documents and implemented the framework for parsing XML documents.

ENVIRONMENT: Python 2.7, Django 1.4, JSON, XML, SQLite, MySQL, HTML5, CSS3, JavaScript, JIRA, GIT, JQuery, Angular JS, Pytest, Windows, Agile.